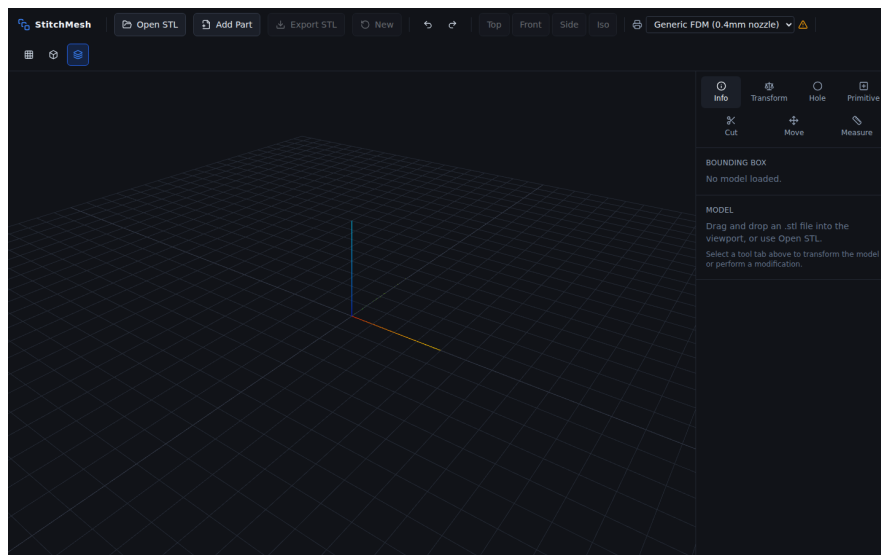




StitchMesh

Complete User Manual

“Modify, don’t model.”

An offline tool for making localized edits to an existing .stl file — resize a hole, add a threaded boss, cut a model in two, measure a clearance — without learning a full CAD package.

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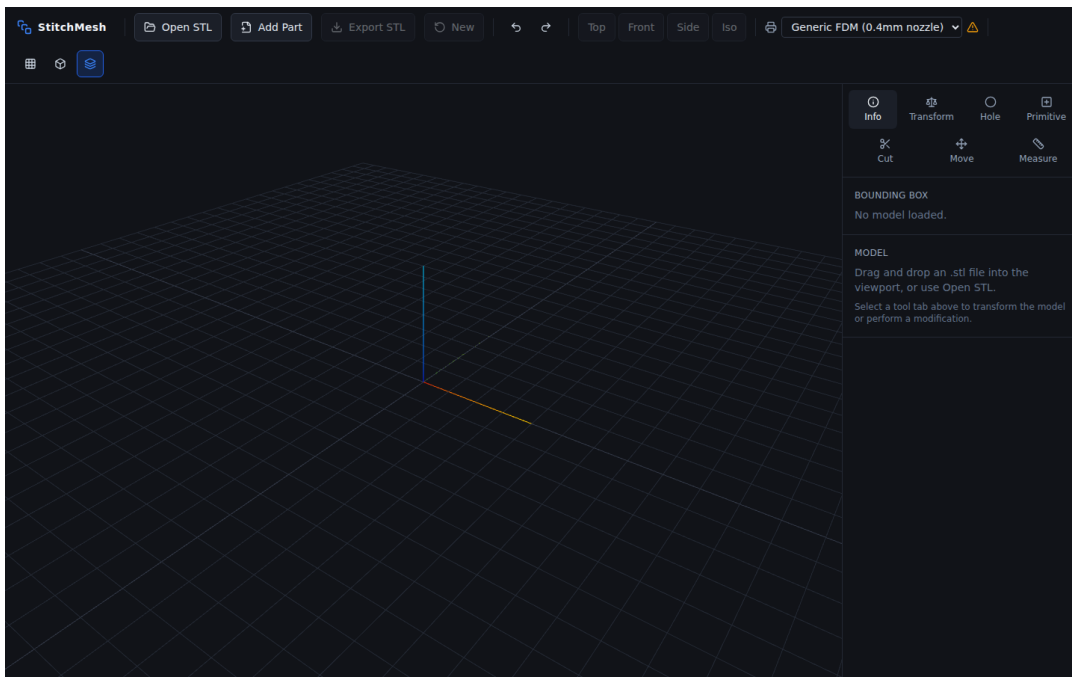
"Modify, don't model." StitchMesh is an offline tool for making localized edits to an existing `.stl` file — resize a hole, add a threaded boss, cut a model in two and move the pieces apart, measure a clearance — without learning a full CAD package.

This document is written for another AI to read alongside a copy of the built app

([dist-standalone/index.html](#)) and produce a tutorial video. Every section names the exact UI element, what it does, what input it expects, and shows a real screenshot of the result. Screenshots live in [docs/manual-assets/](#) next to this file.

1. What StitchMesh is

A three-panel app: a **toolbar** across the top (file, undo/redo, view, printer, display toggles), a **3D viewport** filling the center, and a **context sidebar** on the right with seven tool tabs whose contents change based on which is active: **Info**, **Transform**, **Hole**, **Primitive**, **Cut**, **Move**, **Measure**.



Empty state. Toolbar across the top (Open STL, Add Part, Export STL, New, Undo/Redo, view presets, printer picker, display toggles). Empty dark viewport with a grid and axes gizmo. Sidebar defaulted to Info, reporting "No model loaded."

It ships in three forms, all built from the same source:

Build	Command	What you get
Desktop app	<code>npm run dev (dev) / npm run build (packaged)</code>	Electron app with native Open/Save file dialogs
Web bundle	<code>npm run build:web</code>	Multi-file dist/ folder; needs a static server

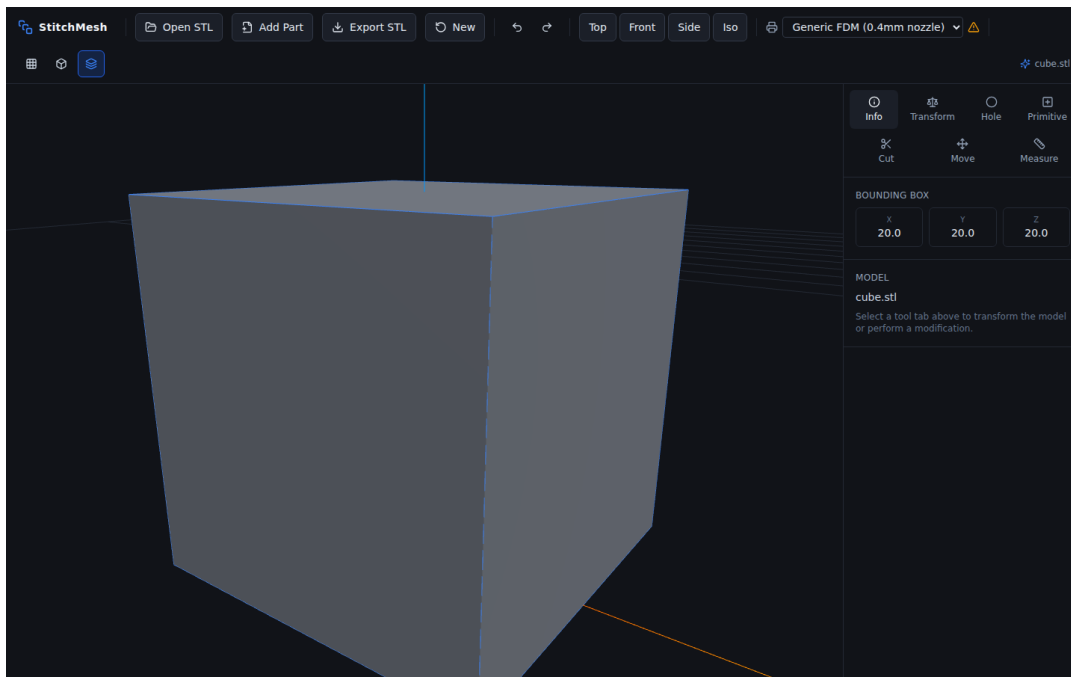
Build	Command	What you get
Standalone	<code>npm run build:standalone</code>	One <code>dist-standalone/index.html</code> file — double-click it, works fully offline, no install, no server. Import/export fall back to drag-and-drop and browser downloads.

All screenshots here were captured from the standalone build running in a real (non-Electron) browser, fully offline.

2. Loading, adding, exporting, and resetting a model

- **Import:** drag an `.stl` onto the viewport, or click **Open STL** (native file picker on desktop, browser picker in standalone). This *replaces* everything currently in the scene. Reading a file never modifies it on disk.
- **Add Part:** imports another `.stl` as a second, independent object next to the first, rather than replacing it — see §11 (Multiple Objects). If nothing is loaded yet, Add Part behaves exactly like Open STL.
- **New:** clears the current model (and its undo history). Asks for confirmation first if a model is loaded.
- **Export STL:** saves the current scene — every part in it — as a new file. Desktop: native Save As dialog. Standalone: a browser download named `<original-name>-modified.stl`. Your source file is never overwritten.

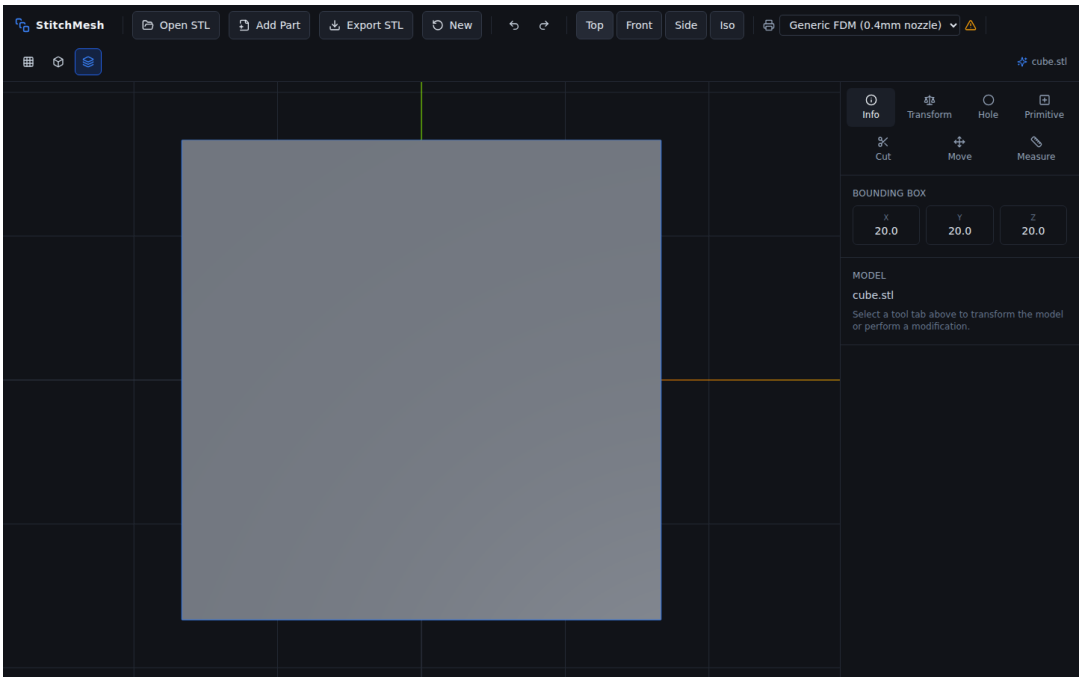
Printer-not-selected export gate: the first time you click Export STL without having picked a printer profile (§14), a dialog interrupts: *"No printer selected yet... Click OK to export anyway with Generic, or Cancel to pick your printer first."* Accepting confirms Generic so it won't ask again.



After import: the model renders, and the Bounding Box readout (visible on every tab) reports live $X \times Y \times Z$ dimensions in millimeters.

3. Viewport navigation

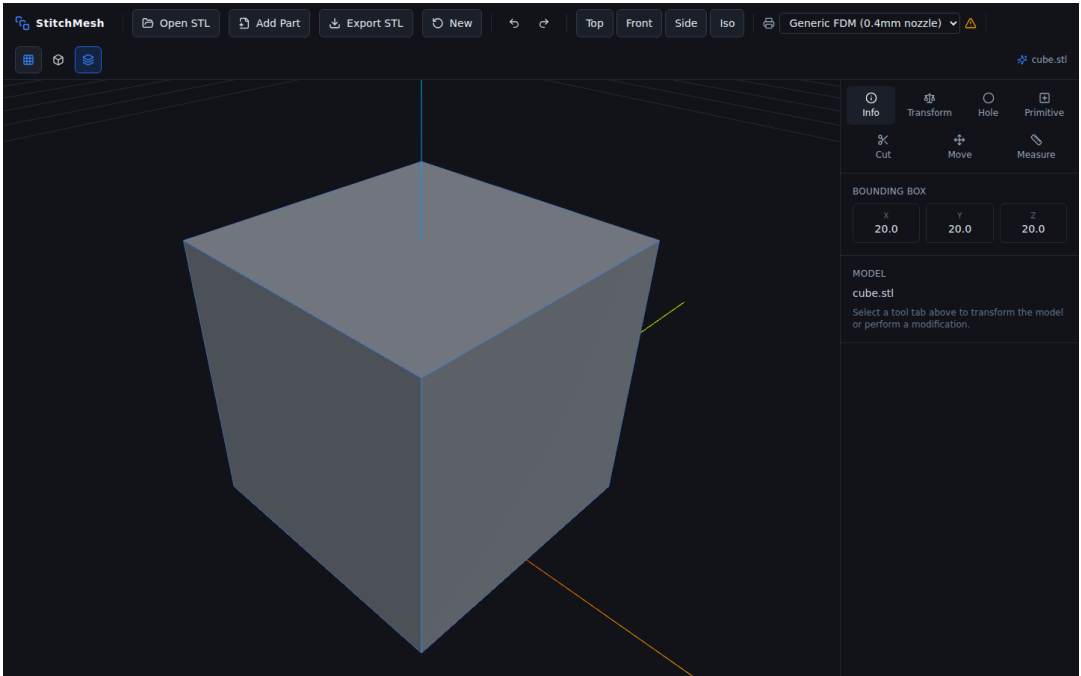
- **Orbit:** left-click-drag, or **middle-click-drag**.
- **Pan:** right-click-drag.
- **Zoom:** scroll wheel.
- **Middle-click-drag re-centers on whatever's under the cursor** the moment you press the button, rather than orbiting around a fixed point — click near a corner and drag, and the view pivots around that corner. Left-drag orbits around the same shared pivot.
- **View presets** (toolbar): **Top / Front / Side / Iso**, or press **1 / 2 / 3 / 4** on the keyboard.



Top preset. Presets re-frame and re-target the camera to the model's **current** bounding box, so they stay useful after scaling, moving parts apart, or cutting.

Display toggles (toolbar icons, or keyboard **W / B**):

Icon / key	Toggle	Effect
Grid icon / W	Wireframe	Renders the model as an edge mesh
Cube icon	Flat shading	Faceted instead of smooth shading — shows actual triangle density
Layers icon / B	Bounding box	Shows/hides the blue outline (on by default)



Wireframe on, from the Iso preset.

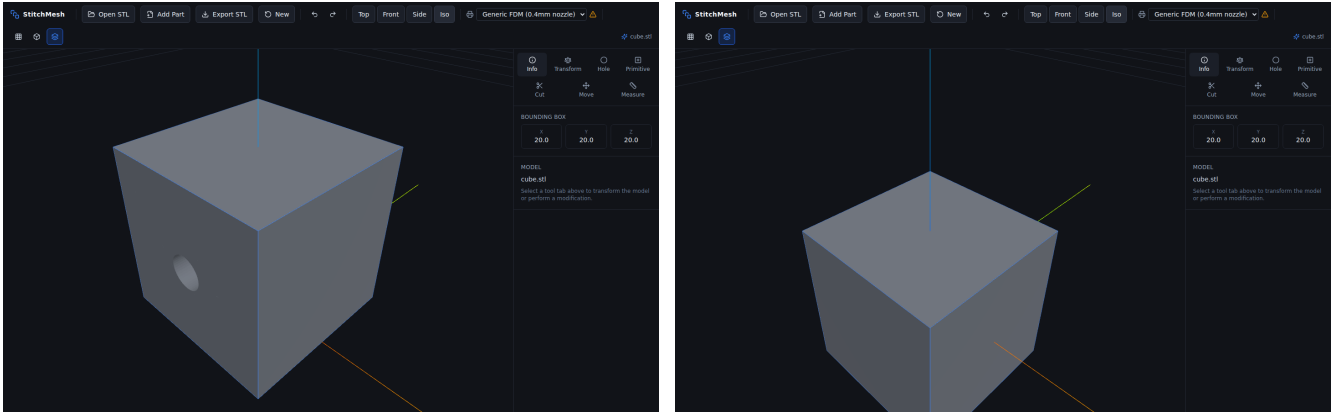
Keyboard shortcuts reference

Key	Action
Ctrl/Cmd + Z	Undo
Ctrl/Cmd + Y, or Ctrl/Cmd + Shift + Z	Redo
1 2 3 4	Top / Front / Side / Iso
W	Toggle wireframe
B	Toggle bounding box

Shortcuts are suppressed while typing in any text field or dropdown.

4. Undo / Redo

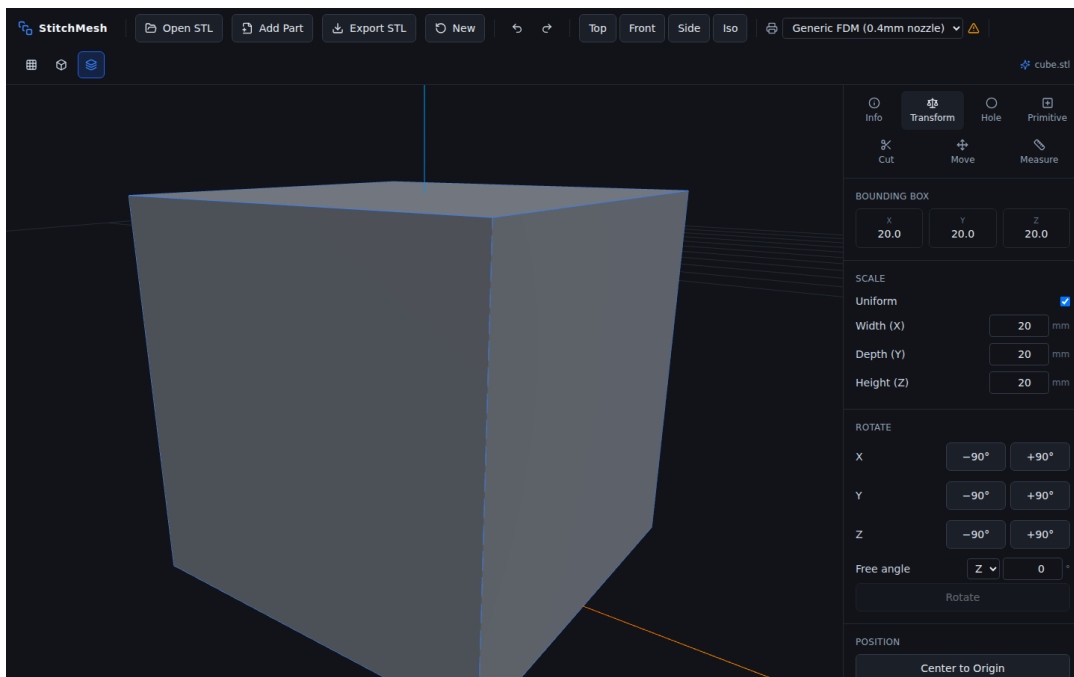
Toolbar: the two curved-arrow icons next to New, or **ctrl+z** / **ctrl+y**. Every committed change is undoable: Hole/Primitive/Plane Cut apply, Transform actions (scale, rotate, center, drop, mirror), Move, adding a part, and mesh repair. Continuous edits (typing digits into a Scale or Move field) are coalesced into a single undo step per edit session rather than one step per keystroke; discrete actions (a button click) are always their own step.



Left: a hole applied. Right: immediately after **Ctrl+Z** — back to the unmodified cube. Redo (button or **Ctrl+Y**) reapplies it.

5. Transform tool

Tab: **Transform**. Affects the whole selected part.



- **Scale** — X/Y/Z fields show the current size in mm; typing a new value rescales to that exact size. **Uniform** (on by default) keeps all three axes proportional when editing one field.
- **Rotate** — $\pm 90^\circ$ snap buttons per axis, plus a **free-angle** row: pick an axis, type any number of degrees, click **Rotate**. Both use the same underlying rotation — the snap buttons are just a fast path for the most common angle.
- **Center to Origin** — recenters the bounding box at world (0,0,0) on all three axes.
- **Drop to Build Plate** — Z-only: shifts so the lowest point sits at Z=0. Runs automatically once on import too.

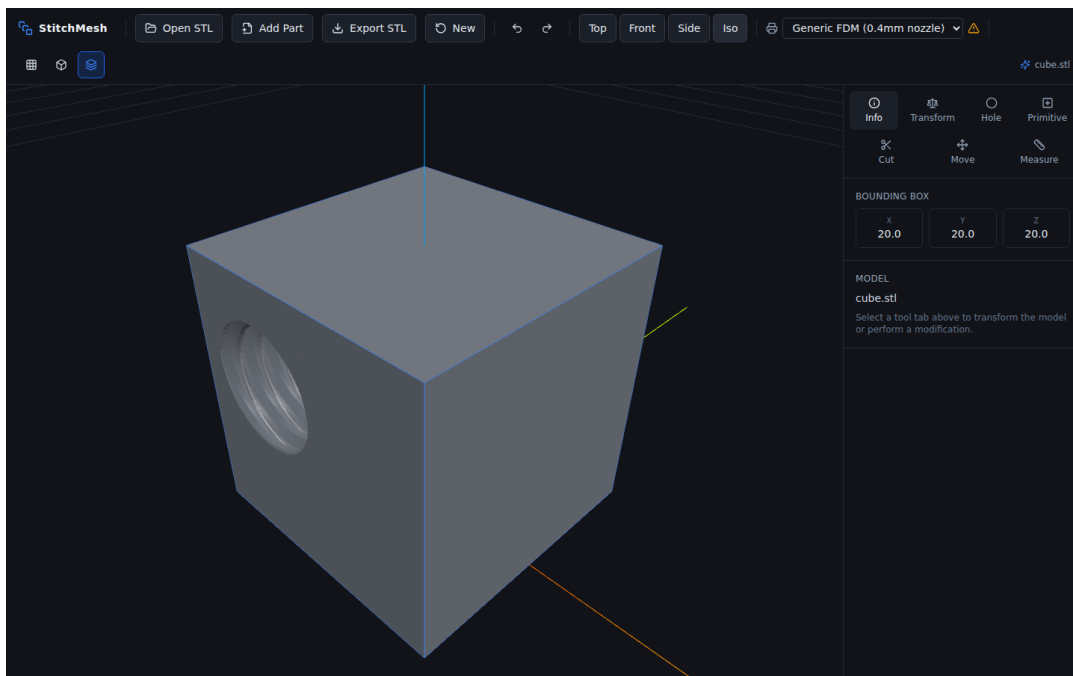
- **Mirror** — flips the part across X, Y, or Z through its own center. Unlike a naive negative-scale mirror (which leaves every face pointing the wrong way), StitchMesh bakes the mirror into the geometry with corrected winding, so raycasting and further edits keep working correctly afterward.

6. Hole Modifier

Tab: **Hole**. Cuts a cylindrical (or threaded) hole at a point you click.

Workflow: click **Hole** → click a point on any part's surface → the sidebar shows Diameter, Depth, Thread, and **Apply Boolean Subtract** / **Cancel**. The part you clicked becomes the target automatically — no extra selection step, even with multiple objects in the scene.

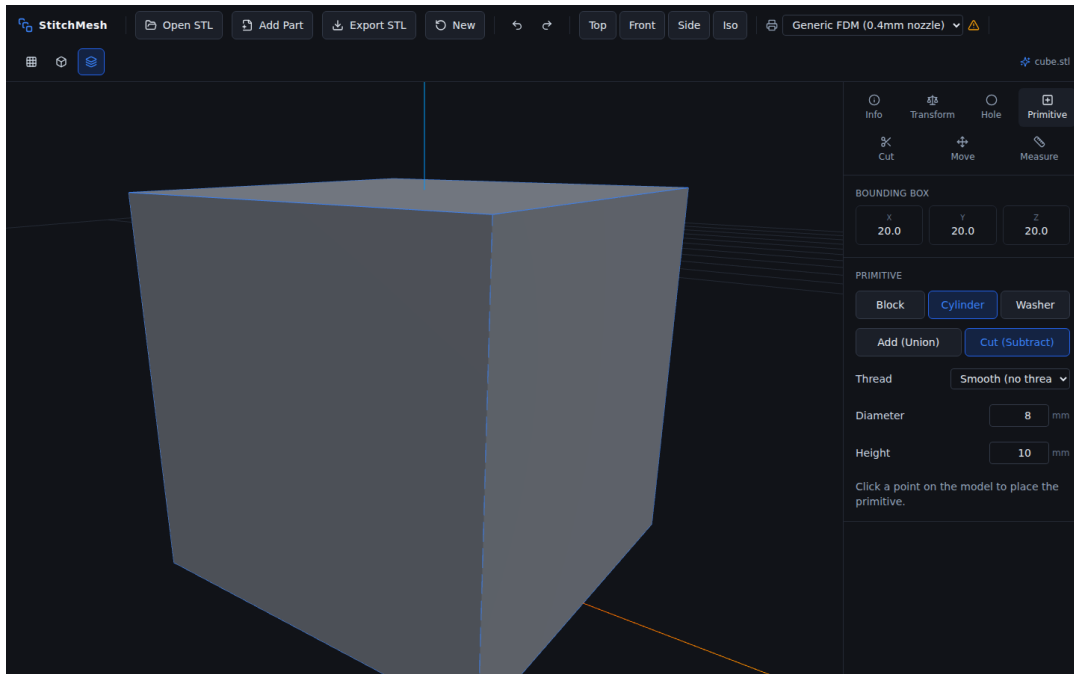
- **Diameter / Depth** (mm) — free-typed when Thread is "Smooth (no thread)". **Depth means depth:** the full stated value is removed from the material, drilling *into* the surface from the clicked point (not split half in/half out).
- **Thread** — a dropdown of standard hardware sizes (§13). Picking one locks Diameter to that thread's exact major diameter and cuts a real helical thread instead of a smooth cylinder.



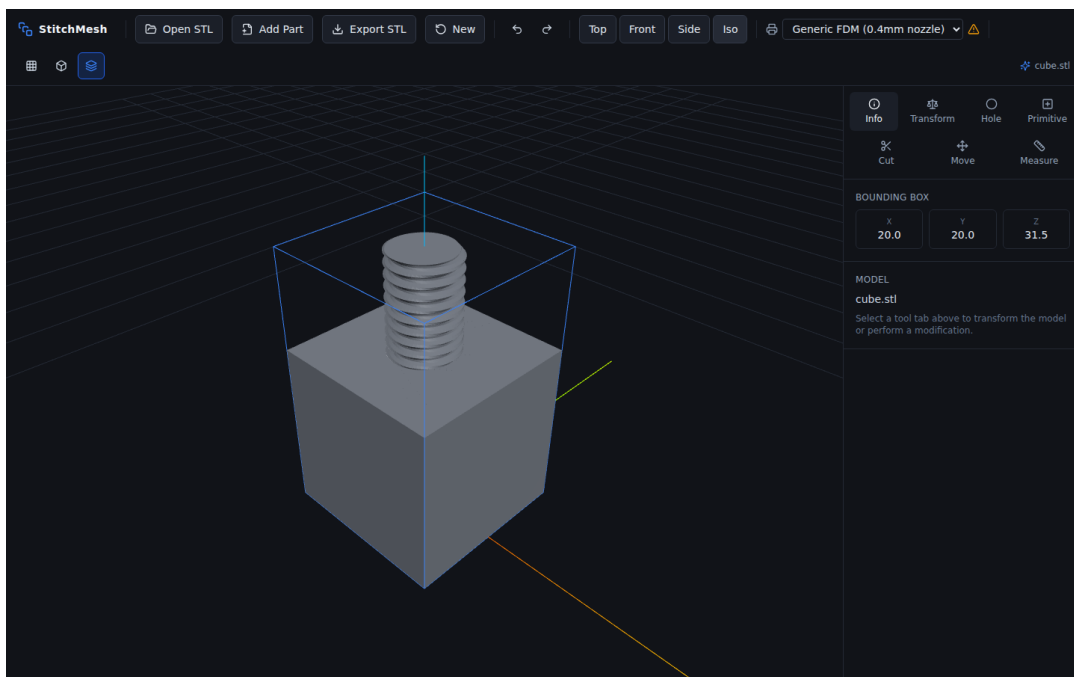
3/8-16 UNC hole applied: a visible internal thread spiraling down the bore, generated from the same geometry a real tap would cut.

7. Primitive Add/Subtract

Tab: **Primitive**. Places a block, cylinder, or washer and unions (adds) or subtracts it from a part at a clicked point — same click-to-place, auto-targeting flow as Hole.



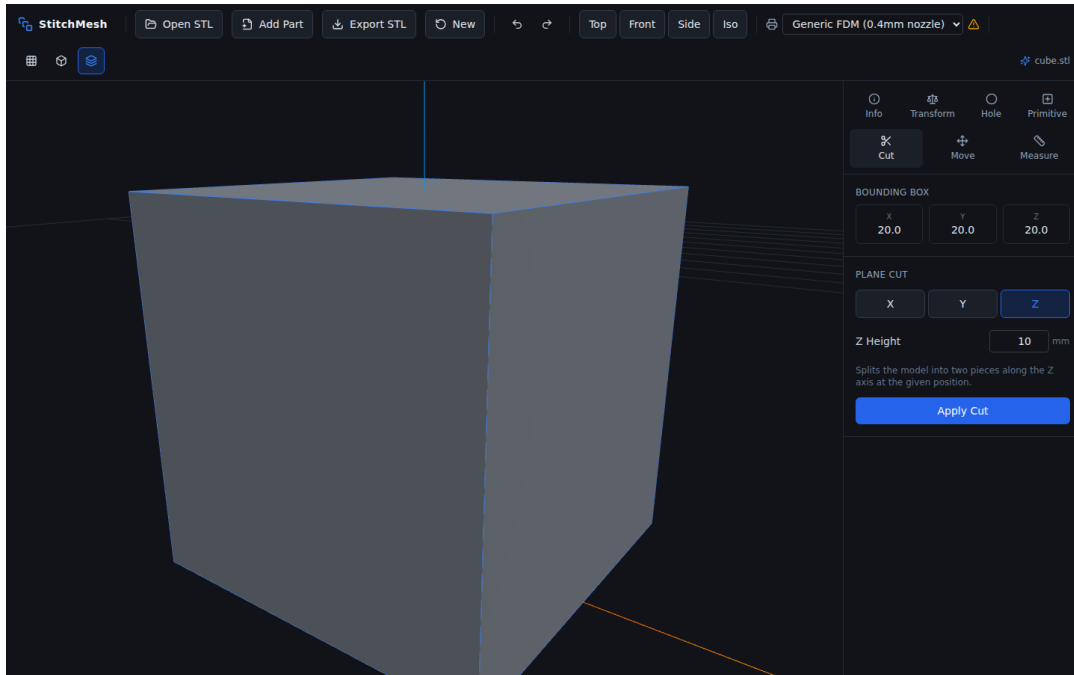
- **Shape:** Block / Cylinder / Washer.
- **Operation:** **Add (Union)** fuses it onto the model as a new protrusion; **Cut (Subtract)** carves it out as a cavity.
- **Dimensions:** Block gets Width/Depth/Height; Cylinder gets Diameter/Height; Washer gets Outer/Inner Diameter/Height.
- **Thread** (Cylinder only): same dropdown as Hole. With **Add** it becomes an external thread (a boss/stud); with **Cut** an internal thread (a tapped hole) — the caption updates to say which.



M8×1.25, Add (Union), 12mm tall, placed on the top face. Bounding-box Z grew from 20.0 to 31.5mm — the full 12mm (minus a deliberate ~0.5mm embed fused into the surface for a clean, non-degenerate boolean).

8. Plane Cut

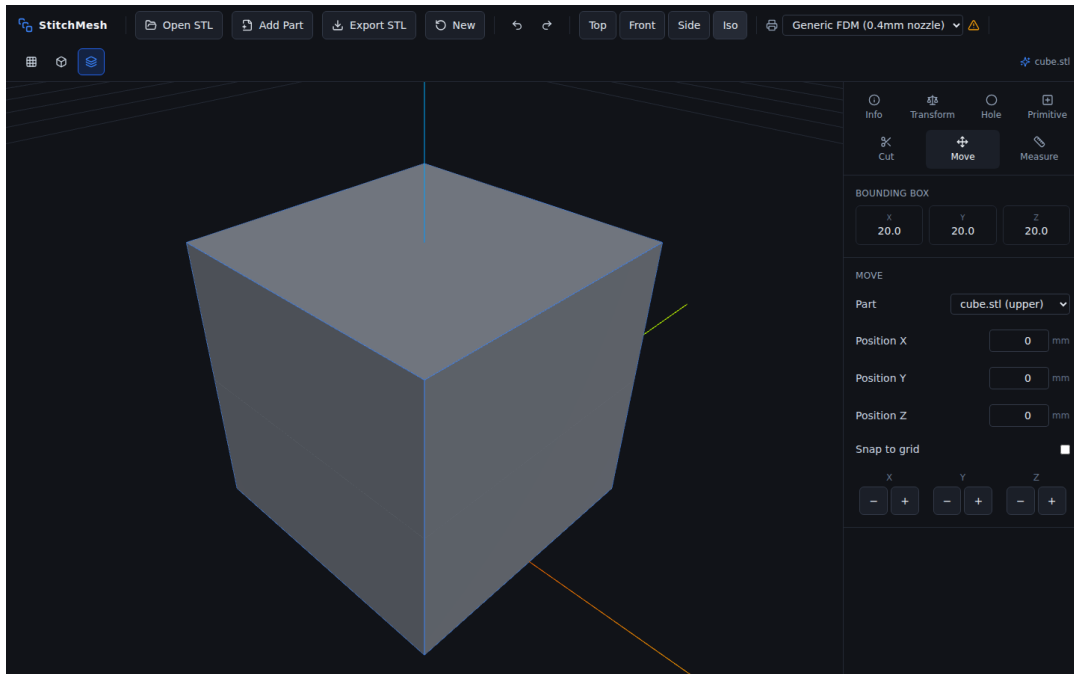
Tab: **Cut**. Splits the target part into **two independent, separately selectable parts** along an axis-aligned plane.



- **Part** selector — only shown once more than one part exists.
 - **Axis** — X / Y / Z, the cutting plane's normal.
 - **Height** — the plane's position on that axis. Auto-defaults to the target's actual center whenever you open the tool or change axis (a model sits on the build plate, so a fixed 0 default would cut at the very bottom edge and remove nothing).
 - **Apply Cut** — both resulting pieces are kept, labeled (**upper**) and (**lower**), and exported together. They start out sitting exactly where the original was — see the Move tool to actually pull them apart.
-

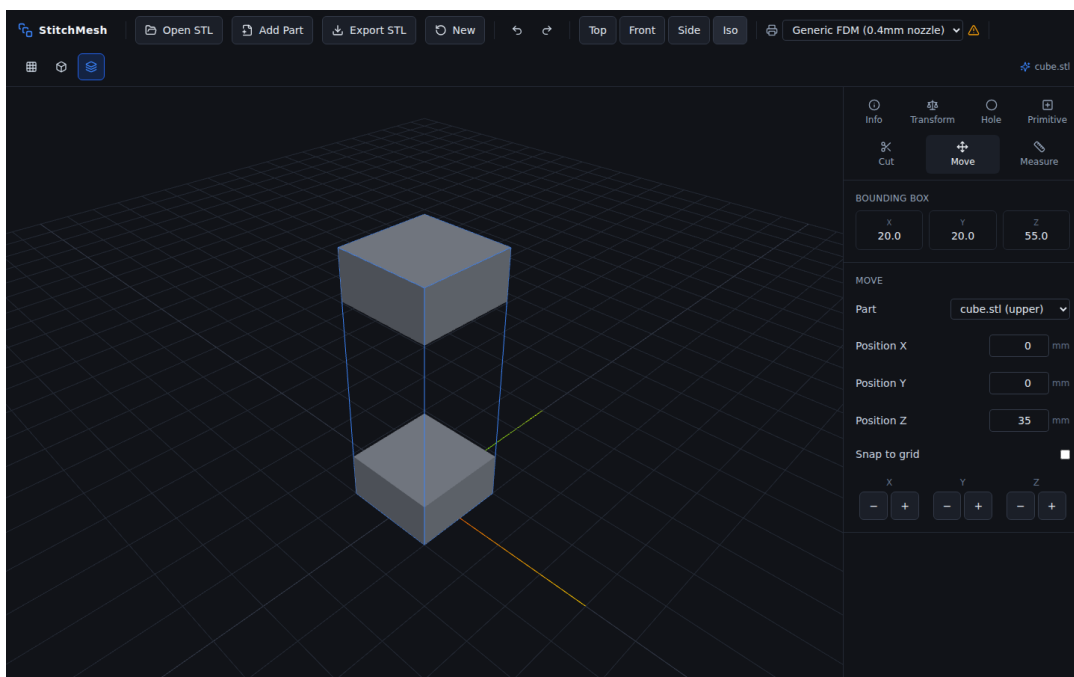
9. Move tool

Tab: **Move**. Selects a part and repositions it — this is what actually separates Plane Cut's two halves (or arranges an added part).



Right after a Plane Cut: two selectable parts exist (`cube.stl (upper)` / (`lower`)), but they still occupy the same space.

- **Part** dropdown — pick which object to move (hidden with only one part).
- **Position X / Y / Z** — absolute world position in mm; typing a value moves the part there directly.
- **Snap to grid** — checkbox; when on, typed positions round to the nearest 1mm.
- **Nudge buttons** (–/+ per axis) — step by the grid size (1mm, or the snap size if enabled).

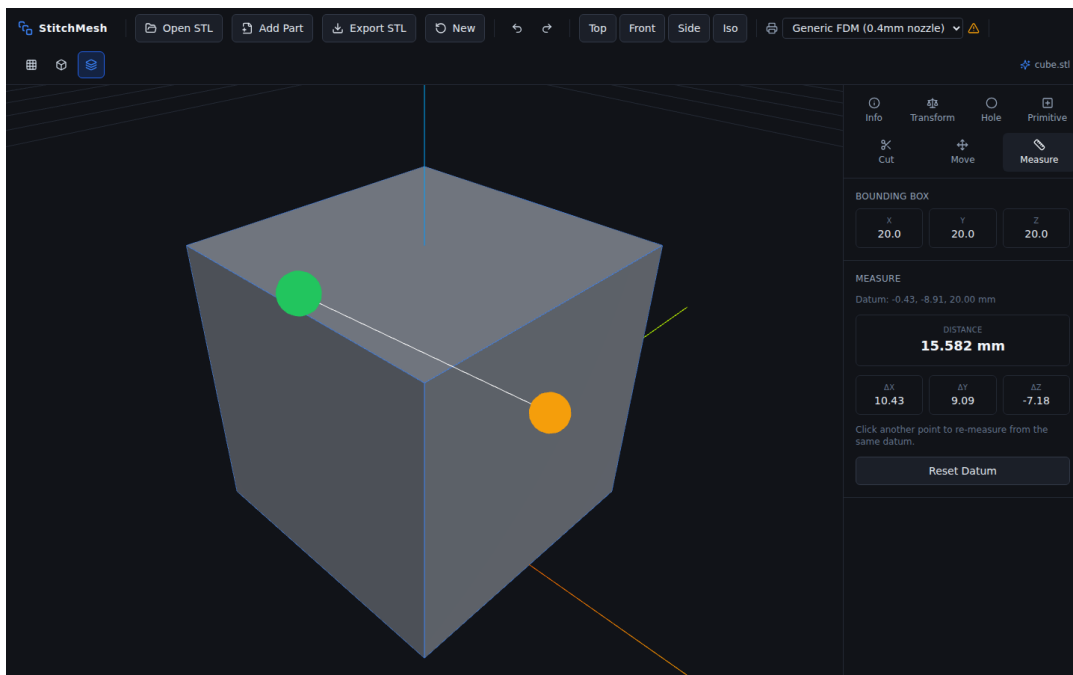


The (upper) piece moved to Z=35: the two halves are now visibly separate, independently selectable, and still export together.

10. Measurement tool

Tab: **Measure**. Click a **datum** (reference point), then click a second point to read the distance and per-axis offset — useful for checking clearances before committing to a hole or thread size.

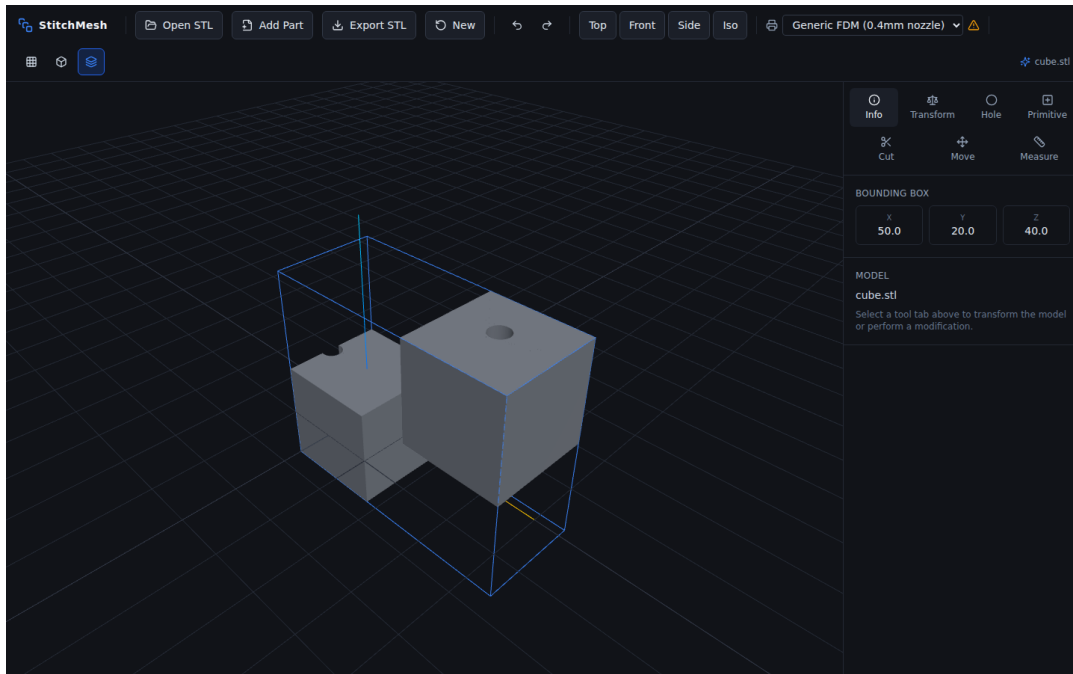
Workflow: click **Measure** → click a point on the model (sets the green datum marker) → click another point (orange marker + a white connecting line) → the panel shows straight-line **Distance** and ΔX / ΔY / ΔZ . Click again anywhere to re-measure from the same datum without resetting it. **Reset Datum** clears both points.



Datum (green) and measured point (orange) on the same cube, with the distance and per-axis deltas shown in the panel.

11. Multiple objects

Add Part (toolbar) imports another `.stl` as an independent object, auto-positioned beside the existing one (offset along X with a 10mm gap, dropped to the build plate) so it doesn't start out overlapping.



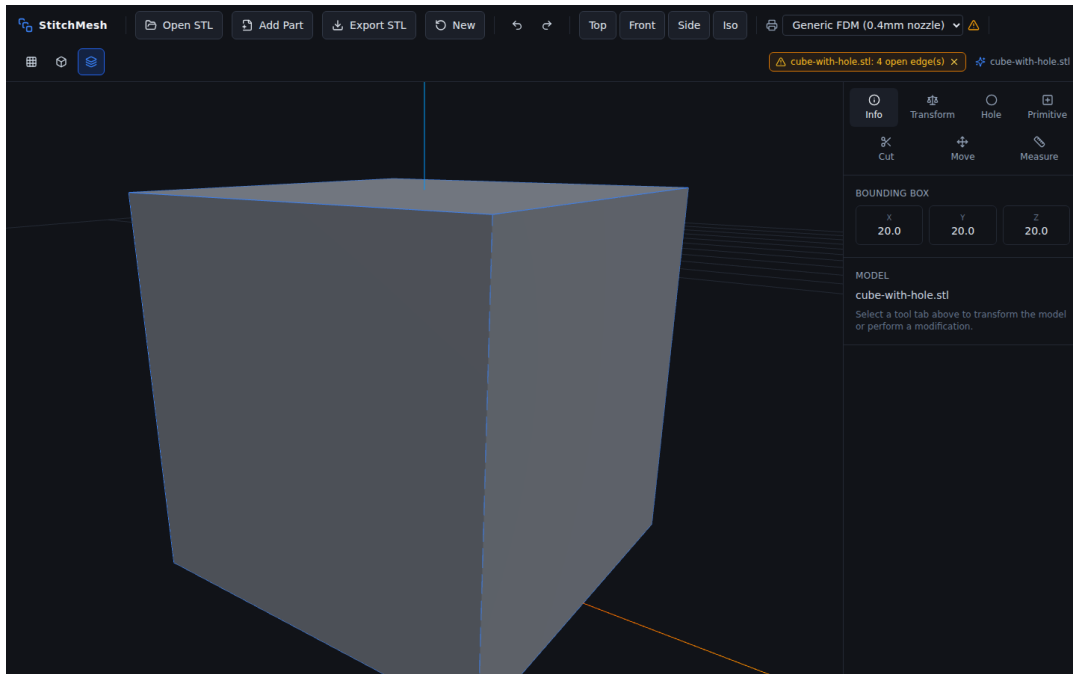
Two cubes in one scene, each drilled independently — clicking a point on either cube automatically targets that part; the other is untouched.

Hole and Primitive always auto-target whichever part you actually click. Plane Cut, Move, and Mirror act on an explicitly selected part (the **Part** dropdown, shown once more than one part exists). Export always includes every part in the scene.

12. Mesh validation & auto-repair

Every import — the first model or an added part — is checked for two kinds of defect and handled differently:

- **Inconsistent or inverted face winding** (a common real-world defect — a few faces flipped, or occasionally an entire mesh reading inside-out) is detected and **automatically repaired**, silently, before you ever see the model. This matters because StitchMesh's click-to-place tools read the *actual* face winding to know which way is "outward" — an unrepaired flipped face would drill in the wrong direction. Repair runs in two passes: a local pass that unifies winding within each connected patch of geometry (by majority vote against its neighbors), then a global pass that flips the whole mesh if it still reads inside-out overall.
- **Open boundary edges** (holes) or **non-manifold edges** (edges shared by three or more triangles — not a valid solid) are **not** auto-fixed, since closing a hole means guessing a shape. Instead, a dismissible banner appears in the toolbar.



A model with one triangle deliberately removed: "cube-with-hole.stl: 4 open edge(s)" with a dismiss (X) button. The banner is informational only — StitchMesh doesn't attempt to fill the hole automatically.

13. Thread generator & standard sizes

Both the Hole and Primitive (cylinder) tools share one **Thread** dropdown with three groups:

- **Metric (ISO):** M2, M2.5, M3, M4, M5, M6, M8, M10, M12, M14, M16, M18, M20 (coarse pitch)
- **Unified Coarse (UNC):** #4-40, #6-32, #8-32, #10-24, 1/4-20, 5/16-18, 3/8-16, 7/16-14, 1/2-13, 5/8-11, 3/4-10
- **Unified Fine (UNF):** #4-48, #6-40, #8-36, #10-32, 1/4-28, 5/16-24, 3/8-24, 7/16-20, 1/2-20, 5/8-18, 3/4-16

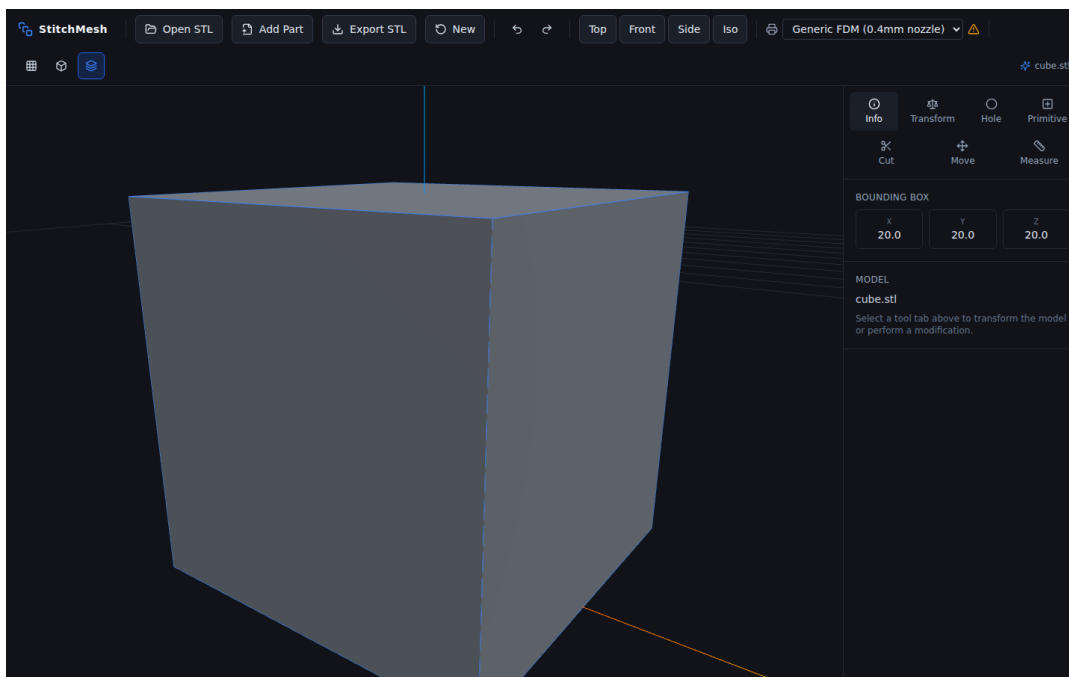
Picking a size generates an actual helical 60° V-thread mesh, from the fundamental-triangle geometry both standards share (ISO 68-1 for metric, ASME B1.1 for Unified):

```
H (fundamental triangle height) = 0.866025 × pitch
pitch diameter                   = major diameter - 0.649519 × pitch
minor diameter                   = major diameter - 1.082532 × pitch
crest truncated by H/8, root truncated by H/4
```

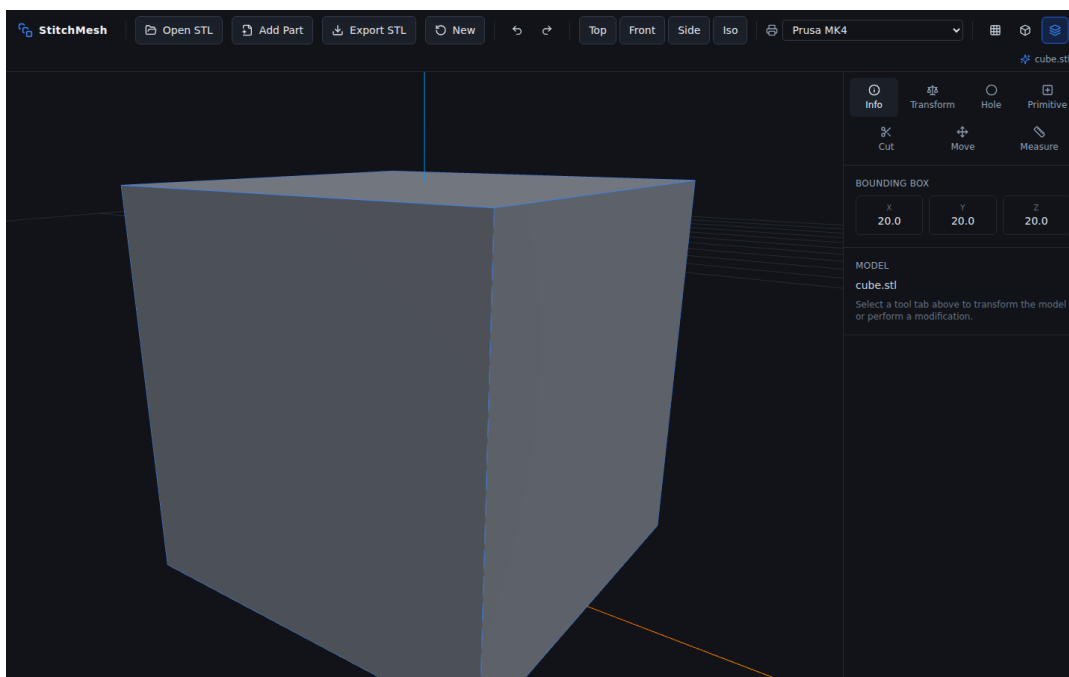
These constants were cross-checked against known published table values (M6×1.0 → minor diameter 4.917mm; M10×1.5 → pitch diameter 9.026mm) — both matched exactly. An internal (tapped) thread and an external (screw) thread are the *same* nominal profile — the generator is reused as a subtraction cutter for one and a union addition for the other, exactly like a real tap cuts the mating shape of the bolt it's sized for. See [src/utils/threadGeometry.ts](#) for the profile math and [src/utils/threadStandards.ts](#) for the size table.

14. Printer profiles

The toolbar's **Printer** dropdown tunes every threaded feature to a specific machine:



Default: "Generic FDM (0.4mm nozzle)" with a warning triangle — shown whenever Generic is active, confirmed or not, since the underlying concern (no machine-specific tuning) is still true either way.



After picking "Prusa MK4": the warning icon clears.

What the profile changes:

- **Mesh resolution** — radial facets and helical height-rings sized to the printer's nozzle/spot diameter and typical layer height.
- **Internal-thread clearance** — a small diametral clearance added to tapped holes (0.25–0.35mm for FDM depending on machine, 0.1mm for resin) so a print accepts a real bolt despite typical over-extrusion.
- **Printability warning** — in the Thread panel, when a pitch is finer than the nozzle can resolve, or the size is small enough (below roughly M4/#8) that a threaded insert would print more reliably.

Included: Generic FDM, Bambu Lab X1 Carbon, Bambu Lab P1S, Prusa MK4, Creality Ender 3 V2, Creality K1C, Voron 2.4 (350mm), Ultimaker S5, Elegoo Saturn 3 (resin), Formlabs Form 4 (resin).

15. Every control, at a glance

Location	Control	Does
Toolbar	Open STL	Import an <code>.stl</code> , replacing the scene (drag-and-drop works too)
Toolbar	Add Part	Import an <code>.stl</code> as a new independent object
Toolbar	Export STL	Save the current scene as a new <code>.stl</code>
Toolbar	New	Clear the model (confirms first)
Toolbar	Undo / Redo	Step back/forward through edit history (<code>Ctrl+Z</code> / <code>Ctrl+Y</code>)
Toolbar	Top / Front / Side / Iso	Jump to a preset camera framing (1–4)
Toolbar	Printer dropdown	Select target printer; tunes thread resolution/clearance
Toolbar	Wireframe / Flat shading / Bounding box	Display toggles (<code>W</code> / <code>—</code> / <code>B</code>)
Sidebar → Info	(read-only)	File name, bounding box
Sidebar → Transform	Scale X/Y/Z, Uniform	Resize the part
Sidebar → Transform	Rotate $\pm 90^\circ$ ($\times 3$ axes), Free angle	Rotate the part
Sidebar → Transform	Center to Origin, Drop to Build Plate	Reposition the part
Sidebar → Transform	Mirror X/Y/Z	Flip the part
Sidebar → Hole	Diameter, Depth, Thread	Configure a hole cutter after clicking a point
Sidebar → Hole	Apply Boolean Subtract / Cancel	Commit or discard
Sidebar → Primitive	Shape, Operation, dimensions, Thread	Configure a primitive after clicking a point
Sidebar → Primitive	Apply / Cancel	Commit or discard
Sidebar → Cut	Part, Axis, Height, Apply Cut	Split the target part in two
Sidebar → Move	Part, Position X/Y/Z, Snap, nudge	Reposition a part

Location	Control	Does
Sidebar → Measure	(click viewport)	Set datum, measure distance/deltas
Viewport	Left/middle-drag, right-drag, scroll	Orbit, pan, zoom

16. Accuracy & validity review

A full-codebase review (an automated correctness pass, plus manual re-derivation of the placement math for every new tool) was run twice this project — once after the initial CSG/thread work, and again after this round's ten new features. All findings below are **already fixed**.

From the first pass:

1. **Hole/Primitive depth was silently halved** — the cutter was centered on the clicked point instead of driven into the surface. Fixed (`positionCutterAtSurface`); confirmed by drilling a hole deeper than the material and verifying it broke through the far face.
2. **Plane Cut's default height (0) was a silent no-op** on any build-plate-dropped model. Fixed to auto-default to the target's actual center.
3. A coordinate-space bug (using each mesh's local transform instead of its world transform in the CSG boolean) that could silently miss entirely after a prior scale/rotate. Fixed; still holds under this round's testing.

From this pass (the ten new features):

- Mesh repair was verified against three synthetic defects: a mesh with two faces deliberately flipped, a fully inside-out mesh, and a mesh with a real open hole. The first two are silently and correctly repaired (confirmed via a through-hole test producing byte-identical triangle counts to a known-good baseline); the third is correctly left alone and reported.
- Mirror was verified to preserve correct winding by drilling a hole immediately afterward and confirming the raycast/CSG pipeline still works — a naive negative-scale mirror would have broken this silently.
- Multi-object targeting was verified by drilling two independent cubes separately and confirming via exported triangle counts that only the clicked part changed each time.
- Undo/redo was verified via both the keyboard shortcut and toolbar buttons, confirming an exact round-trip (12 → 348 → 12 → 348 triangles across apply → undo → redo).
- Snap-to-grid was verified by typing a fractional position (12.6) and confirming it rounded to the nearest 1mm grid line (13).
- Free-angle rotation was verified with a 45° rotation, confirming the resulting bounding box grew to the expected 28.3mm diagonal ($20 \times \sqrt{2} \approx 28.28$).

Everything was re-verified end-to-end in a real (non-Electron) browser, offline, with **zero console errors** across every test. The Electron desktop build was also rebuilt and smoke-tested clean.

Known limitations (not bugs, but worth knowing):

- **No hole-filling.** Mesh repair fixes winding but doesn't attempt to triangulate and close actual holes/non-manifold topology — those are reported, not fixed.

- **Thread mesh triangle counts scale up fast.** A fine-pitch thread over a long depth can generate 10,000+ triangles.
 - **Move and Transform are independent.** Transform (scale/rotate/ center/drop) acts on the whole target part as a group operation; Move repositions one part relative to others. They don't currently share a combined "part-local transform" model — moving a part and then scaling the *whole scene* via Transform will scale every part together, not just the moved one.
 - **Snap-to-grid applies to the Move tool only**, not to Hole/Primitive click placement (snapping a raycast hit point to a grid could shift it off the actual surface).
 - **Single-level undo/redo stack**, capped at 25 steps.
-

17. Possible future features

Most of what was on this list in the previous revision of this manual is now built. What's still genuinely missing, for a future pass:

1. **Hole-filling / non-manifold repair** — closing actual holes, not just fixing winding.
 2. **A true 3D transform gizmo** (drag arrows/rings in the viewport) instead of numeric fields for Move and free rotation.
 3. **Text/label embossing** onto a surface.
 4. **Per-vertex or sculpting-level editing** — StitchMesh is deliberately scoped to primitive-based modification, not freeform mesh editing.
 5. **Saved/named printer profiles** beyond the built-in list (custom nozzle diameter, layer height).
-

Screenshots in this manual live in [docs/manual-assets/](#) alongside this file, captured from the standalone build ([dist-standalone/index.html](#)) at 1440×900 in a real headless Chromium instance, offline.